1–(MAT/11_HL_Summer_2012_Q11)–*MathematicalInduction*

(a) A and U are square matrices, and $X = U^{-1}AU$. Use mathematical induction to prove that $X^n = U^{-1}A^nU$, for $n \in \mathbb{Z}^+$. *[7 marks]*

(b) Let $A = \begin{pmatrix} 2 & -3 \\ 1 & -2 \end{pmatrix}$ and $U = \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}$.

(i) Find the matrix D such that $AU = UD$.

(ii) Write down the matrix D^2.

(iii) **Hence** prove that $A^{2n} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, for $n \in \mathbb{Z}^+$.

(iv) Using the result from part (iii), show that $(A^n)^{-1} = A^n$, for $n \in \mathbb{Z}^+$. *[10 marks]*

2–(MAT/12_HL_Summer_2012_Q13)–*Differentiation,MathematicalInduction*

(a) Using the definition of a derivative as $f'(x)=\lim_{h\to 0}\left(\frac{f(x+h)-f(x)}{h}\right)$, show that the derivative of $\frac{1}{2x+1}$ is $\frac{-2}{(2x+1)^2}$. *[4 marks]*

(b) Prove by induction that the n^{th} derivative of $(2x+1)^{-1}$ is $(-1)^n \frac{2^n n!}{(2x+1)^{n+1}}$. *[9 marks]*

3-(MAT/10_HL_Winter_2012_Q12)**-***Functions-Roots,MathematicalInduction*

Consider a function f, defined by $f(x)=\dfrac{x}{2-x}$ for $0 \le x \le 1$.

(a) Find an expression for $(f \circ f)(x)$. *[3 marks]*

Let $F_n(x)=\dfrac{x}{2^n-(2^n-1)x}$, where $0 \le x \le 1$.

(b) Use mathematical induction to show that for any $n \in \mathbb{Z}^+$

$$\underbrace{(f \circ f \circ \ldots \circ f)}_{n \text{ times}}(x)=F_n(x).$$

[8 marks]

(c) Show that $F_{-n}(x)$ is an expression for the inverse of F_n. *[6 marks]*

(d) (i) State $F_n(0)$ and $F_n(1)$.

(ii) Show that $F_n(x)<x$, given $0<x<1$, $n \in \mathbb{Z}^+$.

(iii) For $n \in \mathbb{Z}^+$, let A_n be the area of the region enclosed by the graph of F_n^{-1}, the x-axis and the line $x=1$. Find the area B_n of the region enclosed by F_n and F_n^{-1} in terms of A_n. *[6 marks]*

4-(MAT/10_HL_Winter_2013_Q6)**-***MathematicalInduction*

[Maximum mark: 7]

Prove by mathematical induction that $n^3 + 11n$ is divisible by 3 for all $n \in \mathbb{Z}^+$.

5-(MAT/10_HL_Winter_2014_Q8)-*MathematicalInduction*

[Maximum mark: 7]

Use mathematical induction to prove that $(2n)! \geq 2^n (n!)^2$, $n \in \mathbb{Z}^+$.

6-(MAT/12_HL_Summer_2015_Q13)**-***MathematicalInduction*

(a) Show that $\frac{1}{\sqrt{n}+\sqrt{n+1}}=\sqrt{n+1}-\sqrt{n}$ where $n \geq 0, n \in \mathbb{Z}$. [2]

(b) Hence show that $\sqrt{2}-1<\frac{1}{\sqrt{2}}$. [2]

(c) Prove, by mathematical induction, that $\sum_{r=1}^{r=n}\frac{1}{\sqrt{r}}>\sqrt{n}$ for $n \geq 2, n \in \mathbb{Z}$. [9]

7–(MAT/10_HL_Winter_2015_Q8)–*MathematicalInduction*

(a) Show that $\sin\left(\theta+\frac{\pi}{2}\right)=\cos\theta$. [1]

(b) Consider $f(x)=\sin(ax)$ where a is a constant. Prove by mathematical induction that $f^{(n)}(x)=a^n\sin\left(ax+\frac{n\pi}{2}\right)$ where $n\in\mathbb{Z}^+$ and $f^{(n)}(x)$ represents the n^{th} derivative of $f(x)$. [7]

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8-(MAT/12_HL_Summer_2016_Q8)**-***MathematicalInduction*

[Maximum mark: 8]

Use mathematical induction to prove that $n(n^2+5)$ is divisible by 6 for $n \in \mathbb{Z}^+$.

9-(MAT/11_HL_Summer_2017_Q8)**-***MathematicalInduction*

[Maximum mark: 6]

Use the method of mathematical induction to prove that $4^n + 15n - 1$ is divisible by 9 for $n \in \mathbb{Z}^+$.

10-(MAT/12_HL_Summer_2017_Q8)**-***MathematicalInduction*

[Maximum mark: 9]

Prove by mathematical induction that $\binom{2}{2}+\binom{3}{2}+\binom{4}{2}+\ldots+\binom{n-1}{2}=\binom{n}{3}$, where $n\in\mathbb{Z}, n\geq 3$.

11–(MAT/10_HL_Winter_2017_Q11)–*MathematicalInduction*

[Maximum mark: 21]

Consider the function $f_n(x) = (\cos 2x)(\cos 4x)\ldots(\cos 2^n x)$, $n \in \mathbb{Z}^+$.

(a) Determine whether f_n is an odd or even function, justifying your answer. [2]

(b) By using mathematical induction, prove that

$$f_n(x) = \frac{\sin 2^{n+1} x}{2^n \sin 2x},\ x \neq \frac{m\pi}{2} \text{ where } m \in \mathbb{Z}.$$ [8]

(c) Hence or otherwise, find an expression for the derivative of $f_n(x)$ with respect to x. [3]

(d) Show that, for $n > 1$, the equation of the tangent to the curve $y = f_n(x)$ at $x = \frac{\pi}{4}$ is $4x - 2y - \pi = 0$. [8]

12-(MAT/11_HL_Summer_2018_Q6)**-***MathematicalInduction*

[Maximum mark: 7]

Use the principle of mathematical induction to prove that

$$1+2\left(\frac{1}{2}\right)+3\left(\frac{1}{2}\right)^2+4\left(\frac{1}{2}\right)^3+\ldots+n\left(\frac{1}{2}\right)^{n-1}=4-\frac{n+2}{2^{n-1}}\text{, where } n\in\mathbb{Z}^+.$$

13-(MAT/10_HL_Winter_2018_Q6)**-***MathematicalInduction*

[Maximum mark: 6]

Use mathematical induction to prove that $\sum_{r=1}^{n} r(r!) = (n+1)! - 1$, for $n \in \mathbb{Z}^{+}$.

14-(MAT/10_HL_Winter_2019_Q6)-*MathematicalInduction*

[Maximum mark: 7]

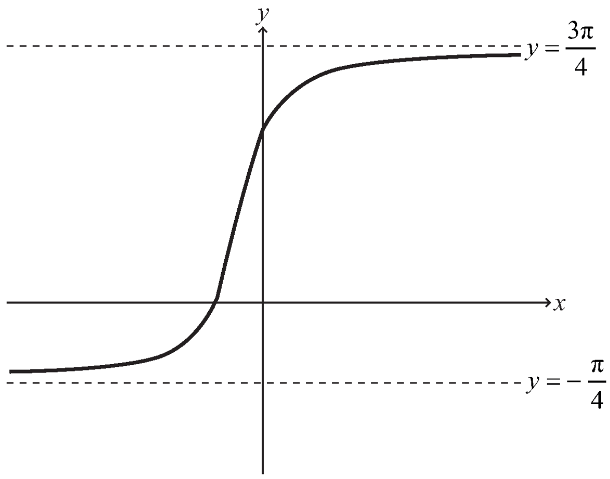
Consider the function $f(x) = xe^{2x}$, where $x \in \mathbb{R}$. The n^{th} derivative of $f(x)$ is denoted by $f^{(n)}(x)$.

Prove, by mathematical induction, that $f^{(n)}(x) = (2^n x + n2^{n-1})e^{2x}$, $n \in \mathbb{Z}^+$.

15-(MathAA/12_HL_Summer_2021_Q12)-*Trigonometry,MathematicalInduction*

[Maximum mark: 19]

The following diagram shows the graph of $y = \arctan(2x+1) + \frac{\pi}{4}$ for $x \in \mathbb{R}$, with asymptotes at $y = -\frac{\pi}{4}$ and $y = \frac{3\pi}{4}$.



(a) Describe a sequence of transformations that transforms the graph of $y = \arctan x$ to the graph of $y = \arctan(2x+1) + \frac{\pi}{4}$ for $x \in \mathbb{R}$. [3]

(b) Show that $\arctan p + \arctan q \equiv \arctan\left(\frac{p+q}{1-pq}\right)$ where $p,\ q > 0$ and $pq < 1$. [4]

(c) Verify that $\arctan(2x+1) = \arctan\left(\frac{x}{x+1}\right) + \frac{\pi}{4}$ for $x \in \mathbb{R},\ x > 0$. [3]

(d) Using mathematical induction and the result from part (b), prove that $\sum_{r=1}^{n} \arctan\left(\frac{1}{2r^2}\right) = \arctan\left(\frac{n}{n+1}\right)$ for $n \in \mathbb{Z}^+$. [9]

16-(MathAA/11_HL_Summer_2021_Q12)-*Differentiation,MathematicalInduction*

[Maximum mark: 20]

Let $f(x)=\sqrt{1+x}$ for $x>-1$.

(a) Show that $f''(x)=-\dfrac{1}{4\sqrt{(1+x)^3}}$. [3]

(b) Use mathematical induction to prove that $f^{(n)}(x)=\left(-\dfrac{1}{4}\right)^{n-1}\dfrac{(2n-3)!}{(n-2)!}(1+x)^{\frac{1}{2}-n}$ for $n\in\mathbb{Z}$, $n\geq 2$. [9]

Let $g(x)=\mathrm{e}^{mx}$, $m\in\mathbb{Q}$.

Consider the function h defined by $h(x)=f(x)\times g(x)$ for $x>-1$.

It is given that the x^2 term in the Maclaurin series for $h(x)$ has a coefficient of $\dfrac{7}{4}$.

(c) Find the possible values of m. [8]